



ARTICLE

TOOL

February 15, 2024

States Can Quantify the Benefits of Climate-Friendly Transportation Options With RMI's Smarter MODES Calculator

RMI's Smarter Mobility Options for Decarbonization, Equity, and Safety (Smarter MODES) Calculator helps decision makers visualize the pollution reduction, cost cutting, and life-saving potential of expanded transportation choices that enable efficient, convenient, and car-free trips.

By Miguel Moravec, Drew Veysey, Bryn Grunwald, Ryan Warsing, and Jackie Lombardi

State Departments of Transportation (DOTs) are positioned to *significantly* reduce transportation pollution, not just with their own programs, but also with the federal funding they alone get to invest.

Historic federal funding is now available through the Infrastructure Investment and Jobs Act (also known as the “Bipartisan Infrastructure Law” or “BIL”). If states spend this funding on expensive, status quo projects like highway expansions, pollution and traffic congestion will rise, health will suffer, and transportation inequalities will persist. If they instead support walking, cycling, public transit, and other clean options, states can reduce these harms and give people of all backgrounds better access to reliable, affordable, and convenient transportation.

Many experts — including at RMI — have shown the necessity of reducing the miles Americans drive (“vehicle miles traveled” or “VMT”), but VMT reduction alone can’t eliminate America’s transportation pollution. In addition to driving less, we need to swap gas-powered cars with electric vehicles (EVs) and build an electrical grid capable of charging them all.

That’s why RMI developed its Smarter Mobility Options for Decarbonization, Equity, and Safety (Smarter MODES) Calculator, which quantifies the benefits of climate-friendly transportation choices in light of customizable EV adoption scenarios. By investing in these choices, **states can avoid billions of dollars in energy, healthcare, and vehicle costs, save lives, and prevent huge amounts of pollution.**

Consumers win big with transportation choices

When people can use non-car modes of transportation, they can save money, find and keep jobs, access quality healthcare and education, and more. The Smarter MODES Calculator shows that with practical auto alternatives, the average American household could save \$750 in annual fuel and vehicle maintenance costs, essentially canceling their annual internet bill. And by putting less wear on their cars, the average household could save over \$1,300 per year in vehicle depreciation. Cumulatively, these household savings could exceed \$8 *trillion* by 2050 — or roughly the value of Microsoft, Apple, and Amazon *combined*.

Average annual household savings from a 20% per-capita VMT reduction in select US states

Results assume a business as usual EV scenario and use state-provided VMT forecasts, where available

Exhibit 1: Average US household savings from a 20 percent per-capita VMT reduction.

The Smarter MODES Calculator takes a wide view of these financial impacts, estimating not only the avoided costs of automobile ownership, but also avoided social costs. For example, by

2050, a 20 percent reduction in national per-capita VMT could save \$408 billion in healthcare costs by improving air quality and \$3.5 trillion by reducing car crashes. States have the power to realize these savings and free these dollars for more productive use.

Americans deserve safer, more equitable mobility options

The monetary costs of car crashes should not detract from their human toll. When cars and people compete for space, cars win. By moving drivers to safer buses and rail transit, and by creating safe, protected routes for cyclists and pedestrians, we lower the likelihood of these deadly collisions and empower Americans to try more active modes of transportation.

Active transportation is healthy transportation: walking and cycling are great ways to stay in shape, and less driving means less pollution that worsens conditions like asthma and heart disease. Moreover, a diversified system with quality public transportation is needed to serve the 28 million Americans who can't afford, can't drive, or simply don't want a car.

Average annual fatalities avoided from a 20% per-capita VMT reduction in select US states

Results assume a mid-growth EV scenario and use state-provided VMT forecasts, where available

Exhibit 2: Average annual fatalities avoided from a 20 percent per-capita VMT reduction.

By 2050, we estimate that a 20 percent reduction in national, per-capita VMT could directly prevent over 180,000 car crash fatalities and indirectly prevent over 1.5 *million* deaths by improving air quality and promoting active lifestyles. That translates to 50,000 lives saved each year — or roughly the capacity of a standard Major League Baseball stadium.

Affordable, clean transportation is needed to meet our climate goals

Each state is responsible for its share of the nation's transportation pollution. As such, the Federal Highway Administration (FHWA) recently unveiled a new rule requiring states to estimate the pollution from federally funded transportation projects — including the Interstate Highway System — and make plans to reduce pollution over time.

Vehicle electrification is an obvious path to doing so. However, it will take some time for all gas-powered vehicles to cycle out of use, and the pollution inherent to cars *in general* —that is,

pollution from sprawl and automotive infrastructure — compels states to consider further measures.

The Smarter MODES Calculator estimates the pollution states could save by promoting EVs *and* lowering VMT. For example, under middle-of-the-road EV adoption projections, paired with a 20 percent decrease in per-capita VMT, we could avoid up to 3 gigatons of pollution by 2050 — the same as turning off *all* emissions, from *all* sectors in the United States, for about six months. Put differently: the right combo of reforms would allow us to meet our climate goals a half-year early, all while unlocking massive societal benefits.

Transportation pollution reduction pathways (gigatons CO₂e)

Exhibit 3: CO₂e avoided (in gigatons) from a mid-growth EV adoption scenario and/or 20 percent per-capita VMT reduction.

Some states are already getting to work. First-movers like California, Colorado, Minnesota, Connecticut, and Washington have taken strong action to reign in excess driving, and similar policies have been considered in New York and Maryland. If the bloc of states in exhibit 2 moved to reduce VMT 20 percent per capita, they could prevent almost an entire gigaton of pollution by 2050 — about a third of the country’s full potential.

By reducing driving, we can make vehicle electrification easier

VMT reduction go hand-in-hand with EVs — and especially with EVs that are not cars: e-scooters, e-bikes, electric golf carts, etc. Compared to electric cars, these lighter EVs require much less energy to operate. When offered as options alongside walking, cycling, and other human-powered modes of transportation, they can help reduce strain on the electrical grid and accelerate the EV transition.

These energy savings are large: in New York state, for example, a 20 percent decrease in per-capita VMT could reduce energy demand by more than 240 Terawatt-hours between now and 2050 — enough to power present-day New York City for over four-and-a-half years. By right-

sizing states’ electricity demand, we can shift utilities’ investments to more critical grid upgrades and install vehicle chargers in more optimal locations.

Everyday Americans would benefit from these utility investments in addition to saving electricity of their own. The Smarter MODES Calculator shows in New York state, a 20 percent decrease in per-capita VMT could save the average household over 1,200 kWh of electricity per year — roughly the consumption of a dishwasher left running for a month.

Exhibit 4: Projected energy savings from a 20 percent per-capita VMT reduction.

States can use the Smarter MODES Calculator to start guiding investments now

An efficient transportation system increases the supply of EVs while decreasing the amount we need to drive. State DOTs can make this system a reality by investing historic funding in climate-smart transportation — like e-bikes and clean buses — instead of conventional highway expansions that predictably increase pollution and impose high social costs.

RMI’s Smarter MODES Calculator can help states calculate the opportunity costs of these status quo investments and motivate a sea change in priorities across all levels of government. These changes will not be easy, but the longer we wait, the more benefits we leave unclaimed.

Related Topics

- SUBNATIONAL ACTION
- TRANSPORTATION
- US POLICY
- UTILITY REGULATION

Authors



Miguel Moravec



Drew Veysey

Senior Associate



Bryn Grunwald

Senior Associate



Ryan Warsing



Jackie Lombardi

Related Insights

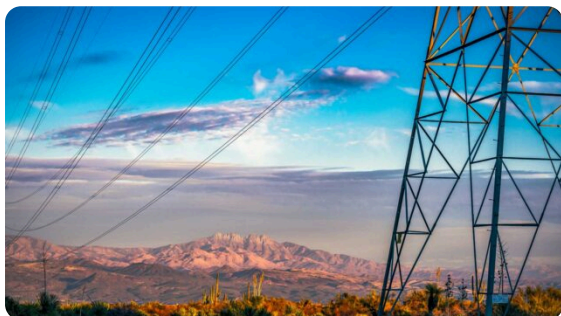


Gas Utility Depreciation: Recommendations for an Uncertain Future

BRIEF

BUILDINGS

September 18, 2026



Subnational Industrial Strategy in the Electro-Industrial Era

BRIEF

US POLICY

September 4, 2026



Turning Local Strengths into Advanced Energy Economies

ARTICLE

US POLICY

September 4, 2026



Sizing and Seizing Economic Opportunity in New Energy Technology

ARTICLE

US POLICY

August 31, 2026

.....





Our Mission

To transform the global energy system to secure a clean, prosperous, zero-carbon future for all.

[Our Approach](#)

[Our Solutions](#)

[Get Involved](#)

[Events](#)

[Careers](#)

[Contact](#)

Sign up for our newsletters

Get regular updates on the global energy transition

Subscribe

